LAB 12 - PROMPTS

Regime-Filter Stage: PCA and K-Means as a Gate on the Signals

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab12-market-regimes/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Build PCA and the regime features (K4)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/regime.py. regime_features(closes) builds backward-looking SPY features: vol_21 (annualised 21-day volatility), ret_21, ret_63, dist_sma200 (close / ta_sma(200) - 1) and corr_spy_tlt (63-day rolling correlation of SPY and TLT returns), dropping NaN rows. fit_regimes(train, k) fits StandardScaler + KMeans(n_clusters=k, n_init=20, random_state=42) and names clusters by average vol_21: 'Low vol', 'Moderate vol', 'High vol (stress)' for k=3, otherwise 'Regime i (vol rank)' with the highest named 'High vol (stress)'. In run() also fit sklearn PCA on standardised daily returns of config universe.trade (pipeline.ingest.load_closes) and print variance explained by PC1-PC4 and the loadings. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: fit_regimes returns a scaler, a fitted KMeans and a dict of names ordered from low to high volatility.

Step 2: Add walk-forward labels and the gate (K6, A4)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/regime.py add walk_forward_regimes(feat, k, first_year=2020): for each calendar year fit_regimes on data up to 31 December of the previous year only and label that year's days; regime_gate(target, regimes, blocked=('High vol (stress)',)) forces the target to 0 on blocked days (the regime is known at the close, so the gate acts on the next bar like any signal). run(k=3) for `python -m pipeline regime --clusters`: print [1] PCA, [2] full-sample cluster profiles (labelled hindsight, description only) with next-day SPY Sharpe and trend-filter Sharpe by regime, [3] walk-forward regime counts and the latest regime and gate state, [4] gated versus ungated CAGR, Sharpe, max drawdown, exposure and trades from 2020 for the configured strategy, trend_filter and dip_buy (backtest with config costs). Save _outputs/Lab12_regime_labels.csv and chart Lab12_regimes.png (PCA variance, SPY coloured by walk-forward regime, gated vs ungated equity). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline regime --help shows --clusters.

Step 3: Run the regime stage and read the PCA (A4, K4)

Do this (no prompt needed)
Run: python -m pipeline regime --clusters 3 (same as: python3 Lab12-market-regimes/regime_stage.py).

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Check: PCA on 9 assets x 2,010 days: PC1 56.8%, PC2 15.5%, PC3 10.3% (cumulative 82.5%). PC1 loads 0.33-0.43 on the equities and -0.09 on TLT: one broad equity factor; TLT and GLD are the diversifiers.

Step 4: Read the regime profiles (K4, K6)

Do this (no prompt needed)
Read section [2]. In which regime does the trend filter work, and where does it fail?

Check: Low vol 654 days (vol 12.7%), Moderate vol 786 days (vol 13.7%, stock-bond correlation -0.33), High vol (stress) 372 days (vol 30.4%). Trend-filter Sharpe 1.23 / 1.04 / -0.45 by regime.

Step 5: Check the walk-forward labels (K6)

Do this (no prompt needed)
Read section [3]. Why does the walk-forward model label far fewer stress days than the full-sample fit?

Check: 2020-2025: Low vol 1,061, Moderate vol 383, High vol (stress) 64 days; latest regime (2025-12-31) Low vol -> gate OPEN.

Step 6: Evaluate the gate after costs (A4)

Prompt - copy into your AI coding assistant
Using the gated versus ungated table from python -m pipeline regime, write a short recommendation to the risk team on whether to switch the regime gate on for ema_cross, trend_filter and dip_buy. Quote Sharpe and max drawdown, mention that the gate is itself a model fitted on few stress days, and say which test from Lab 13 should decide.

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Check: ema_cross 0.91 -> 0.89 and trend_filter 0.89 -> 0.80 (gate hurts); dip_buy 0.63 -> 0.72 with max drawdown -17.98% -> -12.37% (gate helps) - so the gate is signal-specific.

Step 7: Try four clusters (A4)

Do this (no prompt needed)
Run: python -m pipeline regime --clusters 4 and compare the profiles.

Check: Four regimes; the highest-volatility cluster has only 23 days (vol 80.8%, the March 2020 crash) - too few to trust a gate built on it.

Run it

Step	Command / file
Run Lab 12	python3 Lab12-market-regimes/regime_stage.py (same as: python -m pipeline regime)
The complete code	pipeline/regime.py (reference copy: Lab12-market-regimes/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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